Perry Xu

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Education

University of Toronto — Toronto, ON
Bachelor of Applied Science in Engineering Science — *Sept. 2025 – May 2029*
cGPA: 3.92/4.00

Experience

Li Qian Group (Nonlinear & Quantum Optics) | *Research Intern* — March 2026 – Present
University of Toronto — *Toronto, ON*

- Constructed a fiber-based **interferometer** with electro-optic modulation to measure the optical nonlinearity of a **fiber-based nonlinear material**, using a piezo phase modulator to account for environmental phase drift.
- Wrote **Python** scripts to automate the use of the interferometer setup, automatically preparing and extracting complex noisy oscilloscope waveform data through **SCPI**, and processing it to characterize optical nonlinearity.
- Designed and built an STM32-based controller PCB for a high-voltage supply that delivers programmable output over time up to 15kV at 8V resolution ($\sim 0.05\%$ of full scale), enabling precise voltage control in experiments.
- Designed and ran charge dynamics **simulations** to model experimental processes, as well as optical simulations to characterize observed experimental phenomena, using **Python** to automate sweeps and data processing.

University of Toronto Quantum Club | *Team Member* — June 2026 – Present
University of Toronto — *Toronto, ON*

- Built a **PCB-scale NV-diamond vector magnetometer** leveraging optically initialized, microwave-driven spin qubits read out through laser-induced fluorescence; accepted for presentation at **SPIE Photonics West.**
- Designed and simulated a broadband antenna in **Ansys HFSS** to drive NV ground-state spin transitions, holding S11 below -25dB across 2.7-3.0 GHz to cover the full ODMR band around the zero-field splitting frequency.
- Redesigned the board in **Altium** around the new antenna and a lower-phase-noise RF synthesizer to improve achievable spin coherence, and reworked the optical readout system to improve fluorescence collection.

Projects

Optical Power Meter | *Altium, LTspice, C* — February 2026 – March 2026

- Engineered a high-precision optical power meter PCB in **Altium** spanning 80 dB of dynamic range, with layout optimized for signal integrity and a low noise floor across a wide range of light intensities and sampling rates.
- Modeled the analog signal circuitry in **LTspice** with real parasitics and ran noise simulations to predict the noise floor under realistic EMI conditions, then set layout and grounding choices against those results.
- Developed firmware in **C**, featuring an auto-ranging algorithm that dynamically switches op amp feedback paths to maximize range and sensitivity, and times sampling to reduce EMI effects from USB and power traces.
- Programmed a **Python**-based interface program to allow for automatic calibration and optical measurement.

Lumen | *Python, Numerical Modeling* — November 2025 – January 2026

- Engineered a photonic circuit simulator library in **Python** to model, optimize and simulate complex photonic circuits, allowing for validation of experimental setups prior to physical implementation.
- Developed an **efficient numerical solver** for photonic circuits, accounting for nonidealities like losses, crosstalk and phase-shifts across a variety of optical components like interferometers, resonators, beam splitters, etc.
- Designed a modular user interface for defining custom circuit topologies, simulating them in specific ways (e.g., wavelength sweeps) and visualizing the output information in both predefined and custom displays.

Technical Skills

Optics & Photonics: interferometers, optical spectrum analyzers, optical fiber, free-space and fiber alignment, electro-optic modulation, phase modulation, polarization control (polarimeters, controllers), photodetection, lasers
Instrumentation: oscilloscopes, LabVIEW, function generators, RF spectrum analyzers, high-voltage supplies
Programming: Python (NumPy, SciPy, Pandas, PyVISA), MATLAB, C/C++, Qiskit, SCPI, Java
Simulation & Design: COMSOL Multiphysics, Ansys Lumerical, Ansys HFSS, Altium Designer, LTspice

Achievements

IBM Qiskit Fall Fest Hackathon — 2025
3rd Place